

Homotopy types

of schemes

Down under, August 2026

Goal of lecture series. Give a "modern" introduction to étale homotopy theory. Explain recent work refining the étale homotopy type using condensed math.

Some References. Most of what we'll cover is contained in:

- > Hoyois, "Higher Galois theory"
- > H - Holzschuh - Wolf, "The fundamental fiber sequence in étale homotopy theory"
- > H - Holzschuh - Wolf, "Nonabelian basechange theorems & étale homotopy theory"
- > Barwick - Glasman - H, "Exodromy"
- > H - Holzschuh - Lara - Hair - Martini - Wolf, "The condensed homotopy type of a scheme"

# Lecture 1

Notation  $An = \infty$ -category of anima / spaces /  $\infty$ -groupoids

$| \cdot |_{\text{Top}} \rightarrow An$  underlying anima / homotopy type

Desire of étale homotopy theory. Construct an étale homotopy type

$$\Pi_{\infty}^{\text{ét}} \text{Sch} \rightarrow \text{Pro}(An)$$

such that

$$\pi_1 \Pi_{\infty}^{\text{ét}}(X) \simeq \pi_1^{\text{ét}}(X)$$

there are two variants;  
one is a pro-group, the  
other is profinite

$$H^*(\Pi_{\infty}^{\text{ét}}(X), R) \simeq H_{\text{ét}}^*(X, R)$$

discrete ring

and  $\Pi_{\infty}^{\text{ét}}(X)$  "satisfies all of the basic theorems in étale Cohomology"

Problems

(1)  $\pi_1^{\text{ét}}(X)$  is a pro-object

(2) In algebraic geometry, we don't have a reasonable space to do homotopy theory with — we have an  $\infty$ -category of sheaves.

Question that will motivate our definition: Given a reasonable topological space  $T$ , how can we recover the underlying anima  $|T|$  from the  $\infty$ -category  $\mathrm{Sh}(T)$  of sheaves of anima on  $T$ ?

# Shape theory in topology

AKA The nonabelian version of comparing singular and sheaf cohomology

Technical Digression (on hypersheaves). Let  $T$  be a space. Consider the following full subcategories of

$$\mathbf{PSh}(T) = \mathbf{Fun}(\mathbf{Open}(T)^{\mathrm{op}}, \mathbf{An})$$

- (1) The full subcategory  $\mathbf{Sh}(T)$  spanned by the sheaves
- (2) The full subcategory  $\mathbf{Sh}^{\mathrm{hyp}}(T)$  spanned by the presheaves that satisfy descent for hypercoverings
- (3) The localization of  $\mathbf{PSh}(T)$  at the stalk-wise equivalences

Then (2) = (3), and  $\mathbf{Sh}^{\mathrm{hyp}}(T) \subset \mathbf{Sh}(T)$  admits a left exact left adjoint. However, the inclusion is generally strict

Ex. If  $T$  admits a CW structure, then  $\text{Sh}^{\text{hyp}}(T) = \text{Sh}(T)$ .

Basis Lemma. If  $\mathcal{B} \subset \text{Open}(T)$  is a basis for the topology on  $T$ , then the restriction functor

$$\text{Sh}^{\text{hyp}}(T) \longrightarrow \text{Sh}^{\text{hyp}}(\mathcal{B})$$

is an equivalence with inverse right Kan extension along the inclusion  $\mathcal{B}^{\text{op}} \subset \text{Open}(T)^{\text{op}}$ .

van Kampen Theorem. The functor  $|-|: \text{Top} \rightarrow \text{An}$  is a hypercomplete cosheaf, i.e., for every space  $T$  and hypercovering  $U_{\bullet} \rightarrow T$ , the natural map

$$\text{Colim}_{[n] \in \Delta^{\text{op}}} |U_n| \longrightarrow |T|$$

is an equivalence.

Construction. Let  $T$  be a space. Write  $L: \text{PSh}(T) \rightarrow \mathbf{An}$  for the left Kan extension of  $| \cdot |: \text{Open}(T) \rightarrow \mathbf{An}$  along the Yoneda embedding:

$$\begin{array}{ccc} \text{Open}(T) & \xrightarrow{| \cdot |} & \mathbf{An} \\ \downarrow & \nearrow L & \\ \text{PSh}(T) & & \end{array}$$

Then  $L$  has a right adjoint  $R$  given by the formula

$$R(A)(U) = \text{Map}_{\mathbf{An}}(|U|, A).$$

> By the van Kampen theorem,  $R$  factors through  $\text{Sh}^{\text{hyp}}(T)$ .

Prop. If  $T$  is a locally weakly contractible space, then the functor  $L: \text{Sh}^{\text{hyp}}(T) \rightarrow \mathbf{An}$  is left adjoint to the constant hypersheaf functor  $\tau^*: \mathbf{An} \rightarrow \text{Sh}^{\text{hyp}}(T)$ .



Proof.

Write  $\text{Open}_{wc}(T) \subset \text{Open}(T)$  for the subset of weakly contractible opens. Notice that the composite

$$A_n \xrightarrow{R} \text{Sh}^{\text{hyp}}(T) \xrightarrow[\text{rest}]{\sim \text{basis Lem}} \text{Sh}^{\text{hyp}}(\text{Open}_{wc}(T))$$

$$A \mapsto [u \mapsto \underbrace{\text{Map}(u, A)}_{*} \simeq A]$$

is the constant presheaf functor. But it's also a sheaf!  $\square$

Remark. This implies that for a locally weakly contractible space  $T$  and abelian group  $M$ ,

$$C_{\text{Sing}}^{-*}(T; M) \simeq R\Gamma_{\text{sheaf}}(T; M).$$

And the argument is easy!

Upshot. We can recover  $|T|$  by applying the left adjoint to the const. hypersheaf functor to the terminal object.

Idea. In algebraic geometry, we should use the same definition, replacing  $\mathcal{S}h_{hyp}(T)$  with the  $\infty$ -topos of étale sheaves on a scheme.

⚠ The constant étale sheaf functor doesn't admit a left adjoint. But it admits a pro-left adjoint.

# Crash course on $\infty$ -topoi

Thm. Let  $X$  be an  $\infty$ -category. TFAE:

- (1)  $\exists$  a small  $\infty$ -cat  $\mathcal{C}$  and a left exact accessible localization

$$X \begin{array}{c} \xleftarrow{\text{lex}} \\ \perp \\ \xrightarrow{\text{aces}} \end{array} \text{Psh}(\mathcal{C})$$

- (2) Giraud's axioms:  $X$  is presentable and

- (a) Coproducts are disjoint:  $\forall U, V \in X$ , the square

$$\begin{array}{ccc} \emptyset & \longrightarrow & V \\ \downarrow & & \downarrow \\ U & \longrightarrow & U \sqcup V \end{array} \quad \text{is a pullback}$$

- (b) Colimits are universal: for every diagram  $U: A \rightarrow X$ , objects  $V, W \in X$  and morphisms

$$\text{colim}_{\alpha \in A} U_{\alpha} \longrightarrow W \longleftarrow V$$

we have

$$\text{colim}_{\alpha \in A} (U_{\alpha} \times_W V) \xrightarrow{\sim} \left( \text{colim}_{\alpha \in A} U_{\alpha} \right) \times_W V$$

(c) Groupoid objects are effective: if  $U_\bullet: \Delta^{op} \rightarrow \mathcal{X}$  is a simplicial object so that for every partition  $[n] = S \cup T$  with  $S \cap T = \{i\}$  a single point, then

$$\begin{array}{ccc} U_n \simeq U_\bullet([n]) & \longrightarrow & U_\bullet(\{i\}) \\ \downarrow & & \downarrow \\ U_\bullet(S) & \longrightarrow & U_\bullet(\{i\}) \simeq U_0 \end{array} \quad \text{is a pullback,}$$

then  $U_\bullet \simeq \underbrace{\check{\text{Cech nerve of } U_0 \rightarrow |U_\bullet|}}_{\text{Cech nerve of } U_0 \rightarrow |U_\bullet|}$

$$f: X \rightarrow Y \quad \check{\text{Cech nerve is}} \quad \begin{array}{c} \rightrightarrows \\ \rightrightarrows \\ \rightrightarrows \\ \lleftarrow \\ \lleftarrow \\ \lleftarrow \end{array} X \times_Y X \begin{array}{c} \xrightarrow{\text{pr}_1} \\ \xleftarrow{\text{pr}_2} \end{array} X$$

(3)  $\mathcal{X}$  is presentable and colimits in  $\mathcal{X}$  are van Kampen: the functor

$$\begin{array}{ccc} \mathcal{X}^{op} & \longrightarrow & \text{Cat}_{\infty}^{\text{large}} \\ U \mapsto & & \mathcal{X}/U \\ \downarrow & & \uparrow U \times_{\mathcal{X}} (-) \\ V \mapsto & & \mathcal{X}/V \end{array}$$

preserves limits, i.e.,  $\mathcal{X}/\text{colim}_{\alpha \in A} U_\alpha \xrightarrow{\sim} \lim_{\alpha \in A^{op}} \mathcal{X}/U_\alpha$

⚠ Note that we're not saying that there is an  $\infty$ -site  $(\mathcal{C}, \tau)$  and an equivalence  $X \simeq \mathrm{Sh}_{\tau}(\mathcal{C})$ .

This is expected to be false!

Cor (of (3)). If  $X$  is an  $\infty$ -topos, then for any  $u \in X$ , the  $\infty$ -cat  $X/u$  is an  $\infty$ -topos.

Ex.  $T$  Space,  $U \subset T$  open,  $h(U) \in \mathrm{Sh}(T)$  sheaf represented by  $U$ . Then

$$\mathrm{Sh}(T)_{/h(U)} \simeq \mathrm{Sh}(U).$$

Nth. For a scheme  $X$ , we write  $X_{\acute{e}t} = \mathrm{Sh}(\acute{E}t_X, \mathrm{An})$  for the étale  $\infty$ -topos of  $X$ .

Ex.  $X$  scheme,  $U \in \acute{E}t_X$ ,  $h(U) \in X_{\acute{e}t}$  sheaf represented by  $U$ . Then

$$(X_{\acute{e}t})_{/h(U)} \simeq U_{\acute{e}t}.$$

Upshot For a covering  $U_* \rightarrow T$  or étale covering  $U_* \rightarrow X$  we have

$$\mathrm{Sh}(T) \xrightarrow{\sim} \lim_{I \in \Delta} \mathrm{Sh}(U_I) \quad \text{and} \quad X_{\text{ét}} \xrightarrow{\sim} \lim_{I \in \Delta} (U_I)_{\text{ét}}$$

Def. Given  $\infty$ -topoi  $X$  and  $Y$ , a geometric morphism  $f_*: X \rightarrow Y$  is a right adjoint whose left adjoint  $f^*: Y \rightarrow X$  is left exact (= preserves finite limits).

- >  $\mathrm{RTop}_\infty$  =  $\infty$ -topoi and geometric morphisms
- >  $\mathrm{LTop}_\infty$  =  $\infty$ -topoi and left exact left adjoints

$$\begin{array}{c} \text{SI} \\ (\mathrm{RTop}_\infty)^{\mathrm{op}} \end{array}$$

Thm [HTT, Prop. 6.3.2.3 § Cor. 6.3.4.7]  $\mathrm{LTop}_\infty$  has all limits and colimits. Moreover, the forgetful functor

$$\mathrm{LTop}_\infty \rightarrow \mathrm{Cat}_\infty^{\mathrm{large}}$$

preserves limits

## Lecture 2

# Shape Theory

Ntn  $X$   $\infty$ -topos

$\Gamma_* = \text{Map}_X(1_X, -) : X \rightarrow \text{An}$  global sections functor

with left exact left adjoint  $\Gamma^* : \text{An} \rightarrow X$

**Recollection** Let  $\mathcal{C}$  be an  $\infty$ -category. The  $\infty$ -category  $\text{Pro}(\mathcal{C})$  of pro-objects in  $\mathcal{C}$  is an  $\infty$ -category equipped with a functor  $\gamma : \mathcal{C} \hookrightarrow \text{Pro}(\mathcal{C})$  satisfying the following universal property.

(1)  $\text{Pro}(\mathcal{C})$  admits cofiltered limits

(2) If  $D$  is an  $\infty$ -category with cofiltered limits, the restriction functor

$$\text{Fun}^{\text{cofil}}(\text{Pro}(\mathcal{C}), D) \xrightarrow{- \circ \gamma} \text{Fun}(\mathcal{C}, D)$$

↑ cofiltered limit-preserving

is an equivalence.

If  $\mathcal{C}$  has finite limits, then  $\text{Pro}(\mathcal{C})$  has all limits, and  $\gamma$  preserves finite limits.



Observe. For an  $\infty$ -topos  $X$ , Since  $\Gamma^*$  is left exact, the unique cofiltered limit-preserving extension

$$\mathrm{Pro}(A_n) \longrightarrow X$$

of  $\Gamma^*$  preserves all limits. Hence by (an appropriate version of the) adjoint functor theorem, this functor admits a left adjoint

$$\Gamma_{\#} : X \longrightarrow \mathrm{Pro}(A_n)$$

Def.

(1) The shape of an  $\infty$ -topos  $X$  is

$$\Pi_{\infty}(X) = \Gamma_{\#}(1_X)$$

(2) For a Scheme  $X$ , the étale homotopy type of  $X$  is

$$\Pi_{\infty}^{\mathrm{\acute{e}t}}(X) = \Pi_{\infty}(X_{\mathrm{\acute{e}t}}).$$

Ex. If  $T$  is a locally weakly contractible space, then

$$\Pi_{\infty}(\text{Sh}^{\text{hyp}}(T)) \simeq |T|$$

Ex. If  $T$  is the Warsaw Circle, then  $\Pi_{\infty}(\text{Sh}(T)) \simeq S^1$   
as a constant pro-anima

Remark. There is a classical theory of shapes of compact Hausdorff spaces. The shape of  $T$  in the classical sense coincides with  $\Pi_{\infty}(\text{Sh}(T))$ .

Ex. If  $\mathcal{C}$  is a small  $\infty$ -category, then the constant functor  $\Gamma^*: \text{An} \rightarrow \text{Fun}(\mathcal{C}, \text{An})$  admits a genuine left adjoint  $\text{colim}_{\mathcal{C}}$ .  
Hence

$$\Pi_{\infty}(\text{Fun}(\mathcal{C}, \text{An})) \simeq \text{Colim}_{\mathcal{C}} *$$

$$\simeq B\mathcal{C} \quad \text{Classifying anima}$$

$B: \text{Cat}_{\infty} \rightarrow \text{An}$  left adj  
to inclusion.

Q. What is the meaning of the shape?

Obs. For any anima  $A$

$$\mathrm{Map}_{\mathrm{Pro}(\mathrm{An})}(\Gamma_{\#}(1_X), A) \simeq \mathrm{Map}_X(1_X, \Gamma^*(A))$$

Obs. In  $\mathrm{Pro}(\mathrm{An})$

$\simeq \Gamma_* \Gamma^*(A)$  global sections of constant sheaf w/ coefficients in  $A$

$$\mathrm{Map}\left(\varinjlim_i A_i, \varinjlim_j B_j\right) \simeq \varprojlim_j \mathrm{Map}\left(\varinjlim_i A_i, B_j\right),$$

So a proanima is determined by mapping to all anima.

Lem. The functor

$$\begin{array}{ccc} \mathrm{Pro}(\mathrm{An}) & \xrightarrow{\sim} & \mathrm{Fun}^{\mathrm{lex}, \mathrm{accessible}}(\mathrm{An}, \mathrm{An})^{\mathrm{op}} \\ & & \downarrow \text{left exact} \\ A & \xrightarrow{\quad} & \mathrm{Map}_{\mathrm{Pro}(\mathrm{An})}(A, -) \end{array}$$

is an equivalence of  $\infty$ -categories

$$\text{Upshot. } \Pi_{\infty}(X) = \Gamma_* \Gamma^* \in \mathrm{Fun}^{\mathrm{lex}, \mathrm{acc}}(\mathrm{An}, \mathrm{An})^{\mathrm{op}} \simeq \mathrm{Pro}(\mathrm{An})$$

## Functionality of the Shape

Obs The functor

$$A_n \longrightarrow \mathcal{RTop}_\infty$$

$$A \longmapsto \text{Fun}(A, A_n) \simeq A_n/A \quad \text{w/ right Kan extension functionality}$$

is left exact and fully faithful. Hence extends to a right adjoint

$$\lambda: \text{Pro}(A_n) \longrightarrow \mathcal{RTop}_\infty$$

Thm (Lurie). The left adjoint to  $\lambda$  is on objects given by the shape

Corollary. For any  $\infty$ -topos  $X$  and diagram  $U: I \longrightarrow X$ ,

$$\text{colim}_{i \in I} \Pi_\infty(X/U_i) \xrightarrow{\sim} \Pi_\infty(X/\text{colim}_i U_i)$$

Cor. The étale homotopy type

$$\Pi_{\infty}^{\text{ét}} : \text{Sch} \longrightarrow \text{Pro}(\text{An.})$$

is an étale cosheaf.

## Locally Constant objects & monodromy

Def. Let  $X$  be an  $\infty$ -topos. An object  $L \in X$  is locally constant if there exists an effective epimorphism  $\bigsqcup_{i \in I} u_i \rightarrow L$  and anima  $(A_i)_{i \in I}$  such that there exist equivalences

$$\begin{aligned} L \times u_i &\simeq \Gamma_{X/u_i}^*(A_i) && \text{in } X_{/u_i} \\ &\simeq \Gamma_X^*(A) \times u_i \end{aligned}$$

> We say that  $L$  is lisse if, in addition  $I$  can be chosen to be finite and each  $A_i$  can be chosen to be  $[\pi$ -finite].

$\pi_0(A)$  is finite,  $\pi_i(A) = 0$  for  $i \gg 0$  and all  $\pi_i(A)$  are finite

> We write  $X^{\text{lisse}} \subset X^{\text{lc}} \subset X$  for the full subcategories of lisse and locally constant objects

Monodromy v1 (Lurie) Let  $X$  be an  $\infty$ -topos. If  $\Gamma^*: \mathcal{A}_n \rightarrow X$  admits a genuine left adjoint  $\Gamma_\# : X \rightarrow \mathcal{A}_n$ , then the right adjoint to

$$\Gamma_\# : X \longrightarrow \mathcal{A}_n / \Gamma_\#(1_X) \simeq \text{Fun}(\Pi_\infty(X), \mathcal{A}_n)$$

Straightening-unstraightening  
 $\mathcal{A}_n / A \simeq \text{Fun}(A, \mathcal{A}_n)$  + by def.  
 $\Pi_\infty(X) = \Gamma_\#(1_X)$

is fully faithful with image  $X^{lc}$ . That is,

$$X^{lc} \simeq \text{Fun}(\Pi_\infty(X), \mathcal{A}_n).$$

Ex. If  $T$  is a locally weakly contractible space, then

$$LC^{hyp}(T) \simeq \text{Fun}(|T|, \mathcal{A}_n)$$

> For lisse objects, we don't need any assumptions on  $X$ . We first need some technical digressions.

## Background on proanima

Obs. The adjunctions  $A_n \xrightleftharpoons[\perp]{z_{\leq n}} A_{n \leq n}$  induce adjunctions  $\text{Pro}(A_n) \xrightleftharpoons[\perp]{z_{\leq n}} \text{Pro}(A_{n \leq n})$   $n$ -truncated anima:  $\pi_i(A) = 0$  for  $i > n$

⚠ It is possible that a map of proanima  $f: X \rightarrow Y$  induces an equivalence on all  $z_{\leq n}$ , but isn't an equivalence.

If  $X \in A_n$  regarded as a constant proanima and

$$z_{<\infty} A = \{ \cdots \rightarrow z_{\leq n+1} A \rightarrow z_{\leq n} A \rightarrow \cdots \},$$

there is a natural map  $X \rightarrow z_{<\infty} X$  that's an equiv on all  $z_{\leq n}$ . However,

$$\text{Map}_{\text{Pro}(A_n)}(z_{<\infty} A, A) \simeq \text{Colim}_{n \rightarrow \infty} \text{Map}(z_{\leq n} A, A).$$

So every map  $z_{<\infty} A \rightarrow A$  factors through some  $z_{\leq n}$ .

So  $A \rightarrow z_{<\infty} A$  can't have an inverse if  $k$  isn't  $n$ -trunc. for some  $n$ .



Fact The inclusion  $\text{Pro}(A_{n_{<\infty}}) \hookrightarrow \text{Pro}(A_n)$  admits a left adjoint  $\tau_{<\infty}$  that identifies

$$\text{Pro}(A_n) \left[ \left( \text{equivs on } \bigcap_{\text{all } \tau \leq n} \right)^{-1} \right] \xrightarrow{\sim} \text{Pro}(A_{n_{<\infty}})$$

> We call  $\tau_{<\infty}$  the protruncation functor

Def The  $\infty$ -category of profinite anima is  $\text{Pro}(A_{n_\pi})$ . The

inclusion  $\text{Pro}(A_{n_\pi}) \hookrightarrow \text{Pro}(A_n)$  admits a left adjoint

$X \mapsto X_{\pi}^{\wedge}$  called profinite completion.

## Facts

(1) If  $S$  is a set, then  $S_{\pi}^{\wedge} \simeq \beta(S)$  ↖ Čech-Stone compactification

- Since  $\beta$  doesn't preserve finite products,  $(-)_{\pi}^{\wedge}$  generally doesn't either

(2) If  $A$  is a connected anima, then

$$\pi_1(A_{\pi}^{\wedge}) \simeq \pi_1(A) \quad \nwarrow \text{profinite completion of groups}$$

(3) In general, there exist discrete groups  $G$  for which  $(BG)_{\pi}^{\wedge}$  has higher homotopy. In general,

$$(BG)_{\pi}^{\wedge} \longrightarrow B\hat{G}$$

is an equivalence if and only if  $G$  is good in the sense of Serre: for every finite  $G$ -module  $M$ , the natural map

$$H^*(\hat{G}, M) \longrightarrow H^*(G, M)$$

is an isomorphism.

## Monodromy for lisse objects

Ntn Given an  $\infty$ -category  $D$ , write

$$\mathrm{Fun}^{\mathrm{cts}}(-, D) : \mathrm{Pro}(\mathrm{Cat}_{\infty})^{\mathrm{op}} \rightarrow \mathrm{Cat}_{\infty}$$

for the unique colimit-preserving extension of  $\mathrm{Fun}(-, D) : \mathrm{Cat}_{\infty}^{\mathrm{op}} \rightarrow \mathrm{Cat}_{\infty}$ .

$$\mathrm{Fun}^{\mathrm{cts}}\left(\varinjlim_{i \in I} C_i, D\right) = \mathrm{colim}_{i \in I^{\mathrm{op}}} \mathrm{Fun}(C_i, D)$$

Monodromy Thm.  $X$   $\infty$ -topos

There is a natural equivalence

$$X^{\mathrm{lisse}} \simeq \mathrm{Fun}^{\mathrm{cts}}\left(\Pi_{\infty}(X)_{\pi}^{\wedge}, \mathrm{An}_{\pi}\right).$$

Slogan "The profinite shape classifies lisse local systems"

> In fact

Thm (Lurie).  $\lambda : \mathrm{Pro}(\mathrm{An}_{\pi}) \rightarrow \mathrm{RTop}_{\infty}$  is fully faithful with explicit image.

## Basic properties of the étale homotopy type

Convention We'll always protruncate and just write  $\Pi_{\infty}^{\text{ét}}$  for  $\tau_{<\infty} \Pi_{\infty}^{\text{ét}} = \Pi_{<\infty}^{\text{ét}}$

Ex. If  $k$  is a field, then a choice of separable closure  $\bar{k} \supset k$  gives an equivalence

$$\Pi_{\infty}^{\text{ét}}(\text{Spec}(k)) \xrightarrow{\sim} \text{BGal}(\bar{k}/k)$$

Ex.  $\Pi_{\infty}^{\text{ét}}(\text{Spec}(\mathbb{Z})) \simeq *$   $\text{Spec}(\mathbb{Z})$  has no unramified étale covers  $\Rightarrow \pi_1^{\text{ét}} = 0$   
Arithmetic duality  $\Rightarrow H_{\text{ét}}^{\geq 1}(\text{Spec}(\mathbb{Z}); \mathbb{Z}/n) = 0$   
Hurewicz completes the proof

Thm (profiniteness). If  $X$  is a noetherian [geometrically unibranch] scheme, then  $\Pi_{\infty}^{\text{ét}}(X)$  is profinite (e.g. smooth / field)

**Fundamental Fiber Sequence** Let  $X/k$  be a qcqs scheme over a field and  $\bar{k} > k$  a separable closure. There are natural fiber sequences

$$\pi_{\infty}^{\text{ét}}(X_{\bar{k}}) \longrightarrow \pi_{\infty}^{\text{ét}}(X) \longrightarrow \text{BGal}(\bar{k}/k)$$

and

$$\hat{\pi}_{\infty}^{\text{ét}}(X_{\bar{k}}) \longrightarrow \hat{\pi}_{\infty}^{\text{ét}}(X) \longrightarrow \text{BGal}(\bar{k}/k)$$

**Riemann Existence**  $X/\mathbb{C}$  finite type. There is a natural map

$$|X(\mathbb{C})| \longrightarrow \pi_{\infty}^{\text{ét}}(X) \quad \swarrow \text{induced by a geom. mor}$$

$$\text{Sh}(X(\mathbb{C})) \longrightarrow X_{\text{ét}}$$

that becomes an equivalence after profinite completion.

Kunneth formulas.  $k$  field with absolute Galois group  $G_k$ ,  
 $X, Y/k$  qcgs

(1) If  $Y$  is proper/ $k$ ,

$$\hat{\Pi}_{\infty}^{\text{ét}}(X \times_k Y) \xrightarrow{\sim} \hat{\Pi}_{\infty}^{\text{ét}}(X) \times_{BG_k} \hat{\Pi}_{\infty}^{\text{ét}}(Y)$$

(2)  $p = \text{char}(k)$  If  $G_k$  is prime-to- $p$ , then

$$\Pi_{\infty}^{\text{ét}}(X \times_k Y)_{p'}^{\wedge} \xrightarrow{\sim} \Pi_{\infty}^{\text{ét}}(X)_{p'}^{\wedge} \times_{BG_k} \Pi_{\infty}^{\text{ét}}(Y)_{p'}^{\wedge}$$

$(-)_{p'}^{\wedge}$  completion  
 away from  $p$

## Application: Anabelian geometry

Thm (A. Schmidt - Stix).  $k > \mathbb{Q}$  finitely generated extension,  
 $X, Y$  smooth  $k$ -varieties st  $\exists$  embeddings  $X, Y \xrightarrow[\text{closed}]{\text{loc}} \prod \text{hyperbolic curves}$

The the natural map

$$\text{Isom}_k(X, Y) \longrightarrow \pi_0 \text{Equiv}_{BG_k}(\pi_\infty^{\text{ét}}(X), \pi_\infty^{\text{ét}}(Y))$$

is a split injection with a natural retraction

Cor. If  $\pi_\infty^{\text{ét}}(X) \simeq \pi_\infty^{\text{ét}}(Y)$  over  $BG_k$ , then  $X \simeq Y$  as  $k$ -schemes

Thm (Holschbach - J. Schmidt - Stix).  $k$  alg closed field  $\text{char}(k) > 0$ ,  
 $X/k$  smooth  $k$ -variety

Then  $\pi_\infty^{\text{ét}}(X) \simeq *$  if and only if  $X \simeq \text{Spec}(k)$ .

Thm (Achteringer 2017) If  $X$  is an affine connected  $\mathbb{F}_p$ -scheme, then  $\hat{\pi}_{\infty}^{\text{ét}}(X)$  is 1-truncated.

See Holzschuh's talk for more applications.



## Application: The Artin-Tate Pairing

Setup.  $X/\mathbb{F}_q$  smooth geometrically connected surface  $\dim = 2d$

$l \nmid q$  prime

$$Br_{nd}(X) := \frac{Br(X)}{\text{divisible elts}}$$

> M. Artin & Tate defined a pairing

$$Br_{nd}(X)[l^\infty] \times Br_{nd}(X)[l^\infty] \longrightarrow \mathbb{Q}/\mathbb{Z},$$

Really defined on  $H_{\text{et}}^{2d}(X; \mathbb{Q}_l/\mathbb{Z}_l(d))_{nd} \xrightarrow{\sim} H_{\text{et}}^{2d+1}(X; \mathbb{Z}_l(d))_{\text{tors}}$

Conj. (Tate, 1966) The Artin-Tate pairing is alternating

History

1967 § 86 Manin gave counterexamples

1996 Uraibe found mistakes in Manin's work

Thm (Feng, 2017) Tate's conjecture is true

> uses analogies from arithmetic topology

Idea. Because of the fiber sequence

$$\Pi_{\infty}^{\text{ét}}(X_{\overline{\mathbb{F}}_q}) \longrightarrow \Pi_{\infty}^{\text{ét}}(X) \longrightarrow B\hat{\mathbb{Z}} \simeq (S^1)^{\wedge}_{\pi}$$

"IR-dim"

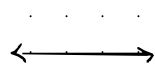
"dim 4"

"dim 5"

"dim 1"

$\Pi_{\infty}^{\text{ét}}(X)$  behaves like a real 5-manifold

Artin-Tate  
pairing



linking form on an  
orientable

$(4d + 1)$ -manifold

> Feng used the étale homotopy type to construct Steenrod operations on étale cohomology + analogies from topology to prove the conjecture

## Lecture 3

Goal Discuss two refinements of the étale homotopy type

(1) To an object (pro-1-category) which classifies constructible étale sheaves

- Barwick-Glasman-H  $\sim 2018$

(2) To a condensed anima whose  $\pi_1$  recovers Bhatt & Scholze's proétale  $\pi_1$

- H.-Holzschuh-Lara-Mair-Martini-Wolf 2025

It turns out These generalizations are intimately related.

Motivation for (2)  $X$  topologically noeth scheme,  $\bar{x} \rightarrow X$  geometric point

> In their work on the proétale topology, Bhatt-Scholze introduced a topological group

$\pi_1^{\text{proét}}(X, \bar{x})$  the proétale fundamental group

With the following key properties:

(1) The profinite completion of  $\pi_1^{\text{proét}}(X)$  is the usual profinite étale fundamental group

- But  $\pi_1^{\text{proét}}(\mathbb{P}_K^1/0 \sim \infty) \cong \mathbb{Z}$

(2) If  $X$  is normal, then  $\pi_1^{\text{proét}}(X) \cong \pi_1^{\text{ét}}(X)$ .

(3)  $\pi_1^{\text{proét}}(X)$  classifies  $\mathbb{Q}_\ell$ -local systems.

(4) In general,  $\pi_1^{\text{proét}}(X)$  is not the limit of a pro-group.

- There is a natural geometric morphism  $X_{\text{proét}} \rightarrow X_{\text{ét}}$  that induces an equivalence on protruncated shapes so  $\pi_1^{\text{proét}}$  can't be accessed via shape theory in the form we've been discussing.

A quick overview of exodromy  
 most simple case of "exit-path" categories in topology

$N_{\text{th}} B. \text{Cat}_{\infty} \rightarrow A_n$  left adjoint to the inclusion

Two ideas. If  $T$  is a top space admitting a triangulation with poset of simplices  $P$ , then

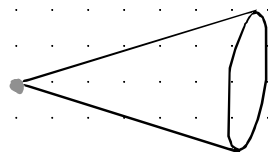
$$(1) \left\{ \begin{array}{l} \text{sheaves on } T \text{ constructible} \\ \text{wrt the triangulation} \end{array} \right\} \xrightarrow{\sim} \text{Fun}(P, A_n)$$

$$(2) |T| \simeq BP$$

General Theory. Exit-path  $\omega$ -categories

> Lurie, Clausen-Jansen, H-Porta-Teyssier

Ex. Constructible sheaves on



are  $\text{Fun}(\bullet \rightarrow \bullet \hookrightarrow \mathbb{Z}, A_n)$

Idea. We should have an algebraic geometry version of this

~ but we'll deal with all stratifications at once

Ntn.  $\text{Cat}_{\infty}^{\pi} = \infty\text{-categories with finitely many objects}$   
up to  $\simeq$  and  $\pi$ -finite mapping anima

$\text{Cat}_{\infty}^{\pi} \cap \text{Cat}_1 = \text{finite 1-categories}$

Def.  $X$   $\infty$ -topos. The  $\infty$ -category of points of  $X$  is

$$\text{Pt}(X) = \text{Fun}^*(X, \text{An}).$$

left exact left adjoints

Recollection  $X$  qcqs Scheme. The category of points  $\text{Pt}(X_{\text{ét}})$  of the étale topos of  $X$  has:

(0) objects: geometric points  $\bar{x} \rightarrow X$

(1) Given geom points  $\bar{s} \rightarrow X$  and  $\bar{\eta} \rightarrow X$ , morphisms are étale specializations

$$\text{Hom}_{\text{Pt}(X_{\text{ét}})}(\bar{s}, \bar{\eta}) = \text{Hom}_Y(\text{Spec}(\mathcal{O}_{X, \eta}^{\text{sh}}), \text{Spec}(\mathcal{O}_{X, s}^{\text{sh}}))$$

> In particular,  $\text{Hom}_{\text{Pt}(X_{\text{ét}})}(\bar{x}, \bar{x}) \simeq \text{Gal}(K(\bar{x})/K(x))$



Thm (étale homotopy type via exodromy, Barwick-Glasman-H.)  
 $X$  qcqs scheme. There is a natural profinite 1-category  $\mathrm{Gal}(X)$  such that

$$(1) \quad \lim_{\leftarrow} \mathrm{Pro}(\mathrm{Cat}_{\infty}) \longrightarrow \mathrm{Cat}_{\infty}$$

$$\mathrm{Gal}(X) \longmapsto \mathrm{Pt}(X_{\mathrm{\acute{e}t}})$$

$$(2) \quad \mathrm{Pro}(\mathrm{Cat}_{\infty}) \xrightarrow{\mathrm{Pro}(B)} \mathrm{Pro}(\mathrm{An}) \xrightarrow{\tau_{<\infty}} \mathrm{Pro}(\mathrm{An}_{<\infty})$$

$$\mathrm{Gal}(X) \longmapsto \Pi_{\infty}^{\mathrm{\acute{e}t}}(X)$$

(3) There's a natural equivalence

$$X_{\mathrm{\acute{e}t}}^{\mathrm{cons}} \simeq \mathrm{Fun}^{\mathrm{cts}}(\mathrm{Gal}(X), \mathrm{An}_{\pi})$$

↳ constructible étale  
 sheaves of anima!

$\exists$  a finite stratification of  $X$  by qcqs  
 subschemes  $Y_i$  such that each  $\mathrm{Fl}_{Y_i}$  is lisse

## Interlude on Condensed math

Setup. The category  $\mathbf{Comp}$  of compact Hausdorff spaces admits a Grothendieck topology where the coverings are finite jointly surjective families

- > Every  $K \in \mathbf{Comp}$  admits a surjection from a projective object.
  - > Gleason showed the projectives are those compacta  $K$  such that for every open  $U \subset K$ ,  $\bar{U}$  is open
- extremally disconnected spaces  $\mathbf{Extr}$
- > On  $\mathbf{Extr}$ , every cover splits, so the basis lemma says

$$\mathbf{Sh}^{\text{hyp}}(\mathbf{Comp}) \xrightarrow{\sim} \mathbf{Sh}^{\text{hyp}}(\mathbf{Extr}) \simeq \mathbf{Fun}^*(\mathbf{Extr}^{\text{op}}, \mathbf{An})$$

$$F(\emptyset) = *$$

$$F(K \sqcup L) \xrightarrow{\sim} F(K) \times F(L)$$

Def.  $\mathcal{C}$   $\infty$ -cat with finite products. The  $\infty$ -category of Condensed objects in  $\mathcal{C}$  is

$$\mathbf{Cond}(\mathcal{C}) := \mathbf{Fun}^*(\mathbf{Extr}^{\text{op}}, \mathcal{C})$$

Fact Consider the functor  $\text{Pro}(\text{Cat}_\infty) \rightarrow \text{Cond}(\text{Cat}_\infty)$  given by extending the const sheaf functor  $\text{Cat}_\infty \rightarrow \text{Cond}(\text{Cat}_\infty)$  along cofilt limits. Then the restriction

$$\text{Pro}(\text{Cat}_{\infty, \pi}) \hookrightarrow \text{Pro}(\text{Cat}_\infty) \rightarrow \text{Cond}(\text{Cat}_\infty)$$

$\hookleftarrow \pi_0(\mathcal{C}^\sim)$  is finite

is fully faithful. and mapping anima are  $\pi$ -finite

Notation Write  $B^{\text{cond}}: \text{Cond}(\text{Cat}_\infty) \rightarrow \text{Cond}(\text{An})$

$$\mathcal{C} \longmapsto [k \mapsto B(\mathcal{C}(k))]$$

for the left adjoint to the inclusion

Def Let  $X$  be a qcqs scheme. The Condensed homotopy type of  $X$  is

$$\Pi_\infty^{\text{cond}}(X) := B^{\text{cond}} \text{Gal}(X)$$

> In particular,  $\Pi_\infty^{\text{cond}}(X)(*) = B(\text{Pt}(X_{\text{ét}}))$ .

Thm. (Wolf). Write  $\text{Cond}(A_n)$  for the condensed  $\omega$ -category

$$K \mapsto \text{Cond}(A_n)_{/A}$$

For a qcqs scheme  $X$ , there is a natural equivalence

$$X_{\text{proét}}^{\text{hyp}} \simeq \text{Fun}^{\text{Cond}}(\text{Gal}(X), \text{Cond}(A_n)).$$

functors of condensed  $\omega$ -categories

Remark. Given a geometric morphism of  $\omega$ -topoi  $f_*: X \rightarrow Y$ , the functor  $f^*: Y \rightarrow X$  always admits a pro-left adjoint  $f_{\#}: X \rightarrow \text{Pro}(Y)$ . Thus one can always talk about the relative Shape

$$\Pi_{\omega}(X/Y) = f_{\#}(1_X)$$

It follows from Wolf's theorem that  $\Pi_{\omega}^{\text{Cond}}(X)$  coincides with the relative Shape of  $X_{\text{proét}}^{\text{hyp}}$  over  $\text{Cond}(A_n)$ .

## Elementary properties/computations

Ex. For a field  $k$ ,  $\Pi_{\infty}^{\text{cond}}(\text{Spec}(k)) \cong BG_k$

> More generally, if  $A$  is a henselian local ring with residue field  $k$ , then

$$\Pi_{\infty}^{\text{cond}}(\text{Spec}(A)) \cong BG_k.$$

Prop (fundamental fiber sequence).  $k$  field with sep closure  $\bar{k}$  and  $X/k$  a gcs scheme. Then the naturally null sequence

$$\Pi_{\infty}^{\text{cond}}(X_{\bar{k}}) \longrightarrow \Pi_{\infty}^{\text{cond}}(X) \longrightarrow BG_k$$

is a fiber sequence.

$\Rightarrow$  long exact sequence on homotopy groups.

Thm. For a compact Hausdorff space  $K$ , there is a natural equivalence

$$\prod_{\infty}^{\text{cond}}(\text{Spec}(C(K, \mathbb{C}))) \simeq K \longleftarrow \begin{array}{l} \text{Condensed set} \\ \text{associated to } K \end{array}$$

Cor.  $\prod_{\infty}^{\text{ét}}(\text{Spec}(C(K, \mathbb{C}))) \underset{\sim}{=} \left( \begin{array}{l} \text{shape} \\ \text{of } \text{Sh}(K) \end{array} \right)$

$$K \text{ locally weakly contractible} \underset{\sim}{=} |K|$$

In particular, for  $K$  LNC,

$$\pi_1^{\text{SGA3}}(\text{Spec}(C(K, \mathbb{C}))) \simeq \pi_1(K)$$

$\pi_1$  of the étale homotopy type, which was studied in SGA3 via different methods

## Comparing fundamental groups

> We have the following issue/feature.

Prop.  $\bar{x} \rightarrow \mathbb{A}_{\mathbb{C}}^1$  any geometric point. Then

$$(\pi_1^{\text{cond}}(\mathbb{A}_{\mathbb{C}}^1, \bar{x})(*))^{\text{ab}} \neq 1$$

Same is true  
for  $\mathbb{P}^1$

In fact, this group has  $\text{card} \geq 2^{2^{\aleph_0}}$ . Hence  $\pi_1^{\text{cond}}(\mathbb{A}_{\mathbb{C}}^1) \neq 1$ .

Def. A condensed set  $T$  is quasicompact if for all epimorphisms  $\bigsqcup_{i \in I} S_i \twoheadrightarrow T$ , there is a finite subcollection that covers

>  $T$  is quasiseparated if  $\forall S \rightarrow T \leftarrow S'$  with  $S$  and  $S'$  qc, the pullback  $S \times_T S'$  is qc.

Ex. If  $T$  is a profinite set, then  $T$  is qcqs as a Condensed Set.

Ex.  $\pi_1^{\text{cond}}(\mathbb{A}_{\mathbb{C}}^1)$  is not quasiseparated

Thm (Clausen-Scholze) The inclusion  $\text{Cond}(\text{Set})^{\text{qs}} \hookrightarrow \text{Cond}(\text{Set})$  admits a left adjoint  $S \mapsto S^{\text{qs}}$  that preserves finite products.   $(-)^{\text{qs}}$  does not extend to Condensed anima

Thm. If  $X$  is a topologically noeth scheme that is geometrically unibranch, then

$$\pi_1^{\text{cond}}(X)^{\text{qs}} \xrightarrow{\sim} \pi_1^{\text{ét}}(X)$$

Fact (Bhatt-Scholze)  $\pi_1^{\text{proét}}(X)$  is a Noohi group:  $G$  Hausdorff group st

$$G \cong \text{Aut}(\text{forget}: \text{Set}^G \rightarrow \text{Set}) \text{ w/ compact open topology}$$

Thm. If  $X$  is a qcqs scheme with finitely many irreducible components, then

$$\pi_1^{\text{cond}}(X, \bar{x})^{\text{Noohi}} \xrightarrow{\sim} \pi_1^{\text{proét}}(X, \bar{x})$$



## Relation to Logic

Setup  $T$  a complete first-order theory,  $\mathcal{U}$  sufficiently saturated model of  $T$

> Lascar defined a quasicompact top group  $\text{Gal}_L(T, \mathcal{U})$

- Up to isomorphism,  $\text{Gal}_L(T, \mathcal{U})$  doesn't depend on  $\mathcal{U}$

> Construction is quite complicated / intricate

Notation  $\text{Mod}_T = \text{Mod}_T(\text{Set})$  Category of models of  $T$

Ex If  $T$  is the theory of (abelian) groups, rings, ..., then  $\text{Mod}_T$  is the category of "

Thm (Cousins - Campion - Ye 2021) There is an isomorphism of discrete groups

$$\text{Gal}_L(T, \mathcal{U}) \cong \pi_1(\text{BMod}_T, \mathcal{U})$$

Observe. The category  $\text{Mod}_T$  has a natural enhancement to a condensed category  $\underline{\text{Mod}}_T$ .

$$K \longmapsto \text{Mod}_T(\text{Sh}(K; \text{Set})).$$

Thm (Darmaj-H-Zhang). There is an isomorphism of condensed groups

$$\text{Gal}_L(T, \mathcal{U}) \simeq \pi_1(\mathcal{B}^{\text{Cond}} \underline{\text{Mod}}_T, \mathcal{U}).$$

Relation to  $\pi_1^{\text{proét}}$ . Actually,  $\underline{\text{Mod}}_T$  is a special case of a construction that makes sense for any  $\infty$ -topos  $\mathcal{X}$ :

$$\underline{\text{Pt}}(\mathcal{X}) \in \text{Cond}(\text{Cat}_{\infty})$$

$$\underline{\text{Pt}}(\mathcal{X})(K) = \text{Fun}^*(\mathcal{X}, \text{Sh}(K)).$$

When  $\mathcal{X}$  is the classifying  $\infty$ -topos of the theory  $T$ , this recovers  $\underline{\text{Mod}}_T$ .

Fact. For a qcqs scheme  $X$ ,  $\text{Gal}(X)$  is naturally a full condensed subcategory of  $\underline{\text{Pt}}(X_{\text{ét}})$ , but  $\text{Gal}(X)$  is much smaller.

Prop(H.). For each extremally disconnected compact Haus space  $K$ , the inclusion

$$\text{Gal}(X)(K) \hookrightarrow \underline{\text{Pt}}(X_{\text{ét}})(K)$$

admits a left adjoint. Hence

$$\mathcal{B}^{\text{cond}} \text{Gal}(X) \xrightarrow{\sim} \mathcal{B}^{\text{cond}} \underline{\text{Pt}}(X_{\text{ét}}).$$

Upshot. Up to Noohi completion,  $\pi_1^{\text{proét}}(X)$  and  $\text{Gal}_L(T)$  are both special cases of

$$\pi_1(\mathcal{B}^{\text{cond}} \underline{\text{Pt}}(X))$$

applied to different topoi.